Chemistry

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===== SOURCE_FILE: nexus sem/Chemistry/PYQs/CT Papers 12-09-2025.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
OSRM
Program: BTech
Course Artieulation Matrix
Course Code & Title: 21CVBI01J & Chemistry Year & Sem: I \Year & 1 Sem
COI
CO2
AL the end of this course, learnrrs ilt he alde tn:
Ratuonalize bulk propaties using periodic properties of clements, evaluate water quality parameters like hardness and alkal1nity
DEPARTMENT OF CIHEMISTRY College of Engincering and Technology SRM Institute Sclence and Technology
Kattankulathur-601201
COS
INTERNAL ASSI SKMENE-IEL
Uilize the concepts of themodynamics in underslanding thcrmouynamically driven chemical rcactions, detenunc acidic strength and redox potentials of' aqueous solution
CO3 Perçeive the importance of stercochemistry in synthesizing organic molecules applied in pharmaceutical industrics, determine acidic strength and conductancc of aqueous
Course Outeomes (CO)
Utilize the concepts of polymer processing for various technological applications, deteminc average imolecular weight of the polymer
(a) [Co(CN)6] (c) [Co(H,0).]
Analyze the importance of advanced processing techniques towards engincering applications and mcasure lhe acidic strength of aqueous solution
(a) 0 A, and -1.2 A,
(c) -1.2 Ao and -1.2 A,
Part -A (5 x |=5 Marks)
(a) [Cu(CN)4]
Answer ALL the Questions
(b) (Co(C;0.b) (dHCo(NH,)6]*
(-1.2 A, and 1.2 A, (d) -1.2 A, and 0 A
3. In spectrochemical series, the order of ligands is
(a) n acceptor < donor < o donor

(c) n donor < o donor < t acceptor
Date: 12-0 92025
2. The CFSE values for d' and d configurations of weak field octahedral complexes are
Duratiem: 12.10-L.29 PM
Ma Marks: 25 marks
3
1
1. Which of the following octahedral complexes of cobalt will have highest magnitude of
Aoet?
(c) [CoBraJ^2
1
3
PO
(bo acceptor <n donor <o donor
(d) n acccptor < o donor <n donor
Set-2
4. Which of the following complexes is paramagnctic (Z of Fe = 26, Co = 27, Cu = 29)
(b) [Fe(CN)Jt
4
2
2
() (Co(NHB)}
3

[Page 2]
3SRM
Program: B. Tech
Course Articulation Matrix
Course Code & Tide: 21CYBIO1J& Chcmistry
Year & Sem: I Year & I Sem
COL
CO2
CO4
Rationalize bulk propertics using periodic properies of elements, cvalunte water
quality parameters like hardncss and alkalinity
AL the end of this course, learners witl be ahle ta:
DEPARTMENT OF CIUEMISTRY
College of Engineering and Technology
SRM Istitute of Science and Technology
Kattankulathur -603203
Ulize the concepts of themmodynamics in understanding themodynamically driven
ical reactions, determine acidic strength and redox potentials of nqucous solution
Perceive the impartance of stercochemistry in synthesizing erganic molccules applied
in pharmnaceutical industries, determine acidic strength and conductance of aqueous
solution

INTERNAL ASSESSMENT-I|FJTL
Course Outenmes (CO)
(a) 0
Unlize the concepts of polymer processing for various technological applications,
Temine average molecular weight of the polymer
Analyze the importance of advanced processing techniques towards engincering applications
and mcasure the acidic strength of aqueous solution
(c) [Ni(CO)4]
(b) I
(b) 2
Part -A (5 x I=5 Marks) Answer ALL the Questions
(c) 2 1. In K«Fe(CN)Js] complex, the number of unpaired clectrons in iron are:
(d) 4
(b) K,[Fe(CN)] (d) [Pi(NH3).]Cl:
RAQ3IONEO O32)
(c) 4.
(b) 0.30
are: T209-20s
Marks: 25 marks
(d) s
3
3
(c) 1.65
3
2
3
3
3. For [Cu(py):(NH;):Clz] complex, the numbcr of geometrical isomers possiblc is (a)
POs
2. Which of the following is a complex compound in which the oxidation state of the metal is zero? (a) K.[Fe(CN)6]
4, The effective nuclear charge experienced by an electron of Helium atom is (a) 0.35
3
(d) 1.70
4
5. Which of the following complcxes is optically active? (The ligand ox = oxalatc). (a) [CoCl,(NH)a(ox)] (b) [CoCI(NH) (ox)]CI (c) [Co(ox)»] (d) [CoCIBr(NH)a(ox)]
2
2
3
Sct-1

5. According to HSAB theory, which of the following has negative coelation with the hardncss?

(a) oxidation state
(c) clcctronegativity
S and 7.
6a. () Compare the crystal filed splitting in octahedral and tetraltedral complexes. IIlustrate the
answer with neat diagraIs. [6 Marks] (i)Calculate spin only magnetic moment and CFSE for the complex K3[Fe(oxalate)s] [4 Marks]
(b) polarizabil1ty
(d) charge density
b (i) How many isormers are possible for thie | Fe(en);Ch;]' complex'? Explain the types of isomers and draw their structures. (cn = ethylencdianine) (6 Marks]
(i) Wite the nanes of two possible coordination geometries for cacl: of coordination numbers [4 Marks]
Q. No
Part -B (2 x 10 = 20 Marks)
BL
Ta. (i) What is meant by effective nuclear charge? Using Slater's rules calculate cffective nuclear
charge of chlorine atom and citloride anion. Explain which one is larger in size and why?
CO
(OR)
b. (i) Definc IISAB principie? Explain the distinction bctwcen hard and soft acids according to
tlis principle and provide cxamples of cach. [6 Marks]
(i) Why is the first ionisation energy of magnesium higher than that of aluminium? [4 Marks]
(OR)
Q No BL
6.
7a1.
CO
[10 Marks]
PO

Part-B(2x10 20 Marles) 6a () Calculate CTSE and spin oniy magnetie moments for the complexes [CoF P and [Co(CN)
(4+4 Marks)
(i) Eor [Ti(H:O)6]" complex, the erystal ield splitting energy is 230 kmol Caleulae its CFSE. (2 Marks)
(ODescribe with examples the coordination and linkage isomerism of coordination compounds
G) Draw the structures of two geometrical isomers of [PiClh[NHS].
(5 Marks] Za () Describe the variation of first electron affinity of elements across a period and down a group
in the periodic table. Explain the trends with suitable reasons.
(5 Marks) (i) Calculate the splitting cnergy of the [Ni(H:0)j² complex with a wavenurnber of

20,000 cm
Determine its Aax and predict the colour of the compound.
(OR)
QNO
b.) Calculate the effective nuclear charge expericnced by a 3d-electron in a bromine atom.
(Zof Br= 35).
BL CO
(OR)
PO
(Gi) Aluminium occurs naturally as an oxide, while cadmium is found as a sulphide. Explain this
difference in occurrence based on the HSAB principle.
O.No
6b.
74.
(4+4 Marks]
(2 Marks)
7b
BL CO
(5 Marlks]
PO
(S Marks]

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===== SOURCE_FILE: nexus sem/Chemistry/PYQs/PYQ CT Papers.url =====
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===== SOURCE_FILE: nexus sem/Chemistry/PYQs/PYQ Dec 2024.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
Reg. No.
Max. Marks: 75Time: 3 hours
Marks BL CO PO
31 1 11.
1 2 1 12.
i 2 1 13.
i 2 1 14.
i 2 2 25.
B) $ZnSO_4$ 1 Zn 11 $CuSO_4$ 1 Cu, E = 1.09 volt
D)Zn 1 ZnSll $CuSO_4$ 1 Cu, E = 1.09 volt
21 2 26.
i 3 2 27.
2 31 48.
at 298 k,
21DF1-21CYB101JPage 1 of 3
Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall
invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.
C)Zn 1 $ZnSO_4$ll $CuSO_4$ 1 Cu,
E = 1.09 volt
PART - A (20 x 1 = 20 Marks)
Answer ALL Questions
The following are state functions EXCEPT
A) enthalpy (H) B) heat (q)
C) internal energy(U) D) entropy (S)
When zinc metal, is attached to the ship's hull made of iron, zinc corrode itself instead of the
hull. The type of corrosion is an example of—
A) Galvanic corrosion B) oxidation corrosion
C) liquid metal corrosion D) pitting corrosion
The crystal field splitting energy for octahedral and tetrahedral complexes is related as
A)At~l/2 Ao B)Ao~2 At
C) Ao ~ 4/9 At D) At ~ 4/9 Ao
B.TECH / M.TECH. (Integrated) DEGREE EXAMINATION, DECEMBER 2024
First Semester
21CYB101J - CHEMISTRY
(For the candidates admitted during the academic year 2024-2025)
Suggest the hard acid from the following ions given:

A)Cu^+ B)Mg^{2+}
C)Cd^{2+} D)Hg^{2+}
According to the convention, the Daniel cell is represented as
A)Zn 1 ZnSO4ll Cu 1 CuSO4 ,
E= 1.09 volt
Which of the following is paramagnetic? [Z for Co= 27, Fe=26, Ni= 28]
A) $[Pt(NH_3)_3Cl]^+$ B) $[Ni(CN)_4]^{2-}$
Q [CoBr]4 2’ D) $[Co(NH_3)_6]^{3+}$
For a high spin d4 octahedral complex the crystal field splitting energy will be
A)-0.6Ao B)-0.8Ao
C)0.6Ao D)0.8Ao
If the solubility product of magnesium hydroxide is 2.00 x 10’11 mol3 dm'9
calculate its solubility in mol dm'3 at that temperature.
A) 1.71 x 104 mol dm'3 B) 1.71 x 10'4mol dm'3
C)2.71 x 10'4mol dm'3 0)3.42 x 10'5mol dm'3
THE HELPERS

[Page 2]
2 3 29.
i 2 3 2
1 2 3 211.
1 2 3 2
21 4 113.
214.
21 4 4
1 2
1 2 5 1
1 2
1 2 5 5
20. i 4 5 5

properties?
A) rubbers
C) fibres
d for first-order spectrum?
A)30A
C)50 A
A) Attack of nucleophile
C) Formation of racemic mixture
B)40A
D)60 A
B) plastics
D) synthetic
B) Electron impact spectroscopy
D) X-ray photoelectron spectroscopy

If X-ray of vvavelength 100 A is incident chi an atom at an angle of OO9, then what should be the value of
Which of the following monomers are unsuitable for condensation polymerization? 1
A) propanoic acid and ethanol B) butane-dioic acid and glycol
C) diamines and dicarboxylic acids D) hydroxy acids
15. The polymer in which steric placements of the substituent are arranged in such a way to give alternate d and 1 cinfigurations, is known as
A) isotactic polymer B) atactic polymer
C) syndio-tactic polymer D) crosslinked polymers
16. Which of the following is a co-polymer?
A) Polythene B) Bakelite
C) PVC D) Polyacrylonitrile
17. In fiber reinforced composites longitudinal strength is mainly influenced by
A) Fiber strength B) Fiber orientation
C) Fiber volume fraction D) Fiber length
18. In storage battery plates the matrix commonly used is
A) Aluminium B) Lead
C) Silver D) Copper
19. Which of the following methods use soft X-rays to eject electrons from inner shell orbitals?
A) Auger electron spectroscopy
C) X-ray crystallography
If our eyes travel in counter clockwise direction from the ligand of highest priority to the ligand of 1
lowest priority, the configuration is
A) S-Configuration B) R-Configuration
C) E-Configuration D)Z-Configuration
10. When a molecule has a plane of symmetry, it will be
A) Optically active B) Both optically active and optically inactive
C) Enantiomer D) Optically inactive
The infinity of intermediate conformations are called
A) Staggered conformations B) Eclipsed conformations
C) Gauche D) Skew
12. Which step in S^jl reaction is a slow rate determining step?
B) Formation of transition state
D)both attack of nucleophile and formation of
racemic mixture
Which of the following polymer type is not classified on the basis of its application and
4 1
5 1
4 1
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Marks BL CO PO

8 2 1 4
8 2 11
8 2 2 1
8 2 2 2
8 3 3 3
3 38 4
8 1 4 1
8 1
8 2 15
8 2 5 3
Marks BL CO PO
15 3
515
b. What is Bragg's Law? Prove nX = 2d sinG. [7]
******

ii) Aluminium occurs in nature as an oxide, whereas Platinum occurs as sulfide. Explain it on the basis of HSAB principle..
22 a. Derive Nemst equation and give its application towards emf determination by potentiometric titration.
21 a. i)Write a short note on spectrochemical series? Explain the impact of ligands in the formation of high and low spin complexes with suitable examples. [6]
PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Questions
PART - B (5 x 8 = 40 Marks)
Answer ALL the Questions
ii) Compare isotactic and syndiotactic polymers
(OR)
b. Explain in detail n and p doping in conducting polymers.
25 a. Explain the various regions in stress-strain graph with a neat diagram.
(OR)
b. Discuss on ceramic and metal matrix composites with examples.
(OR)
b. Explain in detail the mechanism with equations involved in dry corrosion.
23 a. Sketch the potential diagram and discuss in detail the conformational analysis of n-butane.
ii) What is Pairing energy (P)? Give the relation between crystal field splitting in octahedral complexes (Ao) and pairing energy (P). [2]
(OR)
b. i) Write a note on the geometrical isomerism exhibited by octahedral complexes.
(OR)
b. Discuss elements of symmetry with an example for each.
24 a. i) Give the differences between addition and condensation reactions.
2 5
4 2

1 4
26 Discuss in detail Pourbaix diagram of Iron system with a neat sketch.
27 a. What is the principle underlying in XPS? Give the merits and demerits of XPS analysis [8]
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CATEGORY: PYQs
STATUS: ok

[Page 1]
Reg. No. R A2 o o 3 o2o5 ech /M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023 First Semester
21CYB101J -CHEMISTRY (For the candidates admitted from the academic year 2022-2023) Note: Part A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute. Part-B and Part -C should be answered in answer booklet.) (i) Time: 3 Hours Max. Marks: 75
Marks BL, CO PM PART A (20 x 1 = 20Marks) Answer ALL Questions 1. The crystal field splitting encrgy (Ao) is directly proportional to 2
(A) Geometry (C) Coordination number (B) Number of d-Electrons (D) Oxidation state
3 2. The effective nuclear charge realised by is electron of helium atom is (A) 1.00 (C) 1.70 (B) 1.20 (D) 1.65
3 3. The complex [Pt (NH:):Cl;] exhibits (A) Linkage isomerism (C) Geometrical isomerism (B) Coordination isomerism (D) Optical isomerism 4. The spin only magnetic moment value (In bohr magneton units) of Cr(C0)% is (A) (C) 4.90 (B) 2.84 (D) 5.92 0
4 2 5. For a reaction that has an equilibrium constant of 3.2x10, which of the following statement must be true? (A) AH° is negative (C) AG° is negative (B) AG° is positive (D) AS° is positive 2 2 6. For an isolated system, AU = 0, what will be AS? (A) AS>>0 (C) AS S0 (B) AS<0 (D) AS 20 2 7. In the pourbaix diagram, the form of iron that will predominate at pH12 and at potential of 1.86 V is (A) Fe (C) Fe02 (B) Fe2 D) Fe(OH)s 8. Helmholtz function F is given by (A) -U+ TS (C) U+ TS (B) -U-TS (D) U-TS 05JF21CYBI01J Page 1 of 3

[Page 2]
OF EN cadem1 2 9. The number of structural isomers for CsH14 is (B) 5 3 rst 40 (A) 6 (C)4 (D) ooklet. I2 3 2 10. Reactivity order of alkyl halides in Ss reaction is (A) CH3x1°>2°>3 (C) 32>1°>CH3x (B) CH3x>2°>3°>1° (D) 3>1°>2°>CHx 332 11. Among the following hex-2-ene reacts fastest with? (A) HCI (C) HI in (B) HF (D) HBr 12. Which of the following has the lowest priority according to the CIP 3 3 2 sequence rules? (A) CH(OH) CH; (C) CHO
lutio. (B) CH CH: (D) CH2 CH cult 41 13. Which of the following is a thermo setting polymer? (A) Bakelite C) PVC (B) Polystyrene D) Polyethene
2 4 1 14. Which one of the below is used as an insulator and also as a lubricant? (A) PVC C) SBR
ma (B) PTFE (D) Poly propylene
4 15. Hemodialysis tubes are made with (A) Silicone rubber (C) Polyurethane intermediate s

(B) Polystyrene (D) Nylon
4 16. Which of the below polymers show higher crystallinity? (A) Isotactic (C) Random (B) Atactic (D) Syndiotactic acj
251 17. In fibre reinforced composites which constituent will fail last? (A) Filler (C) Both fail at same time (B) Matrix (D) Need more details on composite
5 18. After the proportionality limit in the stress-strain curve, we observe (A) Lower yield point (C) Ultimate point (B) Upper yield point (D) Elastic point
25 19. Minimum inter planar spacing required for Bragg's diffraction is (A) 4 (C) /2 (B) 4 (D) 2. 20. Determine young's modulus of a material whose elastic stress and strain are 4 Nm and 0.15 respectively (A) 26.66 N/m2 (C) 266.6 N/ms (B) 2.666 N/m? (D) 2666 N/m2

[Page 3]
PART-B (5 x8 = 40 Marks) Answer ALL Questions Marks i. CO PM
21. a. Find the number of unpaired electrons in strong and weak octahedral field 4 ra Mn*" complex (d) based on CFT. Caiculate CFSE and magnetic noment for both the situation with energy level diagrams. (OR) 3 b. Demonstrate with proper examples the isomerism exhibited in transition metal complexes.
22 22. a. With appropriate examples, elucidate how Nemst equation can be appiied in a redox reaction and in an acid-base reaction. (OR) b. Derive Gibbs-Helmholtz equation and given its applications. 2
2. 32 23. a. Compare and contrast Sx' and Ss reactions with an example for cach (OR) b. Sketch the potential energy conformational analysis of n-butane. 3 2 diagram and explain in detail the
24. a. Provide a conscise note on the synthesis and applications of Teflon and PVC. (OR 4 b. Explain in detail n and p-doping in conducting polymers. 35 25. a. Illustrate with a proper stress-strain plot for the following i) (ii) Elastic region Plastic region (OR) b. Explain with an example ceramic matrix composite and metal matrix composite. 25
PART C(1 x 15 15 Marks) Answer ANY ONE Questions Marks BI CO PO
15 26. With an neat sketch discuss pourbaix diagram for iron. 21.1. Explain E2 mechanism with suitable example.
10 i. Discuss about the principle and instrumentation of X-ray photo electron spectroscopy.
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[Page 1]
31. a Derive Gibbs-Helmhotz equation. Give its applications-
(oR)
b. With a neat sketch explain Pourbaix diagram for kon.
32. a Explain in detail the conformational analysis of n-butane with potential erer5/
diagam.
, Reg. No.
B.Tech. DEGREE EXAMINATION, NO\EMBER 2018
First Semester
l8CYBlOIJ - CHEMSTRY
(For the candidates admitted during the academic year 20lB-2019)
Note:
..(l) Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should
be hmded
over to hall invigilator at the end of 45fr minute.
(ii) Part - B and Part - C should be answered in answer booklet.
Time: Three Hours
PART-A (20 x 1= 20 Marks)
AnswerALL Questions
l. Two electrons occupying the sarne orbital me distinguished by
b. How is isomerism
example each?
(oR)
exhibited in transition metal

rirl.***
compounds? Explain the types with an
(A) Azimuthal quantum number
(C) Magnetic quantum number
2. Ttrc de-Broglie hypothesis is associated with
(A) Wave nature of electrons only (B) Wave nature ofprotons only
(C) Wave nature of radiation @) Wave nanre of all material particles
3. For a homonuclear diatomic niolecule the bonding orbital is
(A) orof lowest enerry
(C) arof lowest enerry
(A) Arenes
(C) Acyls
(A) NearIR
(C) Visible regron
(B) Spin quantum number
@) Orbital quantum number
@) o,of second lowest enerry
@) z,of lowest ener5/
(B) Aryls
@) Benzenes
(B) A, =l l2yo
(D) Ao = 419L,
(B) I
(D) 3
@) Microwave region
(D) Radio frequency region
Mo<. Marks: 100
.IMIFUSCYB1O1J
4. Organic compounds which contain more than one benzene rings are terrned
5. The crystal field splitting enersr for ockrhedral (4) and tetrahedral (A,) complexes is related as
(A) L,=4l9Lo
(C) Ao =24,
6. The number of unpaired electons in d6, low spin octahedral complex is
(A) 0
(c) 2
7. The vibrational rotational spectnrm is obseried in _ region
8. In a rotational spectrum transitions are only observed between rotational levels of A.I =
- (s) t2
@) tr
(A) *t(c) 0
Pagc d of4 l7Nr118CrBl0lJ Page 1 of4

[Page 2]
The kinetic eners/ of the ejected photoelectron is dependent upo-n the energr of the

tal I"* **i" @) Photons around
tcj urroia (D) ImPingingPhoton
Compute the miller indices for the intercepts * = yZ,y = /rona' - L
(A) 110
(c) 100
The smallest interplanar spacing in a ctystal which will give ntr order Bragg's reflection is
i;;-i;r=" (B) dnr=%
(C) dna:% @) a* -/o
12. The second ionization enerry is always higher than the firstionization enerry because the
(A) Elector ir Xtru"t"a mire by t1i corJ @) Electron is more tightly bound to the
electrons nucleusinanion
(C) Stability increases on attaining an @) Atomic size is larger
octet or duPlet configuration
The Er mechanism proceeds via formation of
(A) Carbanion (B) Carbocation
(Ci PorUte Uona @) Free radical
Reduction of ketone to hydroxyl group takes place by one of the following reagents'
9.
19.
20.
10. (A) NaBII4
(C) KIvInO4
(B) Oz
@) CrOr
PART-B (5 x 4=20 Marks)
Answer ANY FM Questions
11.
(B) 101
(D) 210
(B) LU -q+w
@) M=Aq+w
2t.
,,",
23.
)4.
25.
Discuss the radial wave functions of hydrogen atom'
Discuss the enerry levet diagram of Oz molecule'
Explainbriefly about high spin and low spin complexes with examples'
Discuss the criteda for absorption in the IR region'
Write a note on the variatioa in atomic and ionic sizes across the periods and groups'
26- Define: Entropy. Give its sigrrificance'
27. Briefly explain Dieckmann condensation'

ici 2." (D) R€srins consd Alsw.t AII Qu€sti@s
15. The name of the equation showing rylation between electrode potential @), standard
potential ("E) urd concentration of ions in solution is
(A) Kohlrausch's law @) Nemst equation
(C) Faraday's equation (D) Ohm's law
16. Corrosion of metals involves
(A) Electrochemical reactions @) Chemical reactions
iC; fomaanan (D) Therrralreactions
13. First law of thermodynamics states that
(A) LU=q-w
(C) LU=q+Lw
17. Enantiomers are
(A) Molecules that have a mirror image
(C) Non-super imposable molecules
Y
H&CH3
H& CzIIs
Molecules ttrat have atleast one stereogenic
center
Non-super imposable molecules that are
mirror images of each other
CH: & CHr
BothB & C
IMIF,T8CYB1O1J
. 28. a Discuss in detail the schrodinger wave equation of a particle in a box'
(oR)
b. what is Linear combination of atomic or6itasr Draw and explain the molecular orbital
energi level diagram for hydrogen molecular ion [r{i] and calculate the bond ordet'
18. In the Newmann projection of 2,2- dimethyl butane x & Y can be represented as
29. a.i. Give the salient features of crystal field theory'
ii. Discuss thd crysul freld splitting in a octahedral complex
(oR)
b. Discuss the vibrational spectrum of a diatomic
motion
molecule undergoing simple harmonic
30. a--Discuss the principle, instumentation and applications of)(PS'
(oR)
Write a note on Vander Waal's interactions' Marks)
Discuss in detail Bragg's law forthe diftaction of crystals with aneat sketch' (s Marks)
lTNFrl8CYBtolJ
(4 Marks)
(8 Marks)
(B)
(D)
(B)

(D)
(A)
(c)
b.i.
ii.
Prge 2 ofl
Pogc 3 of4

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STATUS: ok

[Page 1]
Reg. No.
Max.,Marks: 75Time: 3 hours
Marks BL CO PO
1 21.
1 2 1 4 •2.
1 2 1 4 .3.
i 2 1 44.
2 • 2 31
1 2 2 36.
2 31 27.
2 2 .318.
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Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall
invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.
B) AS surroundings>0 only
D) AS system>0 only
PART - A (20 x 1 = 20 Marks)
Answer ALL Questions
B.TECH DEGREE EXAMINATION, NOVEMBER 2024
First and Second Semester
21CYB101J - CHEMISTRY
(For the candidates admittedfrom the academic year 2021-2022 to 2023-2024)
A)-0.6Ao
C)-1.6 Ao+P
Which factor is primarily responsible for the difference in magnetic properties between high-spin
and iow-spin octahedral complexes?
A) Ligand field strength B) Coordination number - , -

C) Oxidation state of the metal D)Nature of the counter-ion
According to the Hard and. Soft Acids and Bases (HSAB) principle, which of the following pairs will
have the strongest ionic interaction?
A) Soft acid and hard base B) Hard acid and soft base
C) Soft acid and soft base D)Hard acid and hard base
The second ionisation energy is always higher than thefirst ionization energy because the.
A)electron is attracted more by the coreB)electron is more tightly bound to the
electrons nucleus in an ion
C) becomes more stable attaining the D) atomic radii is large
octet or duplet configuration
The CFSE for a high spin d4 octahedral complex is.
B)-L8A0
D)-1.2A0
1 4
5. Which of the following is the correct criterion for a spontaneous process?
A) AS system-AS surroundings
C) AS system+AS surroundings >0
The correct statement about cell potential is.
A) sum of the electrode potentials of the B) difference between the electrode potentials
cathode and anode of the cathode and anode
C)half of the sum of the electrode D) twice the difference between the electrode
potentials of the cathode and anode potentials of the cathode and anode
-1
In Pourbaix diagram, the redox reaction, Fe2++2e“->Fe(s) is.
A) solvent dependent B) solvent independent
C) pH dependent D) pH independent
Which corrosion product is volatile in nature?
A)MoO3 B)Fe2O3
C)Fe3O4 D)CuCO3
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3 3 419.
1 3 3 4
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1 2 5 5
1 2 5 5
51 2 5type of material.
i 2 5 5

Marks BL CO PO
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8 4
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PART - B (5 x 8 = 40 Marks)
Answer ALL the Questions
21 a. Describe the salient features of Crystal Field Theory (CFT) and write its limitations.
(OR)
b.How does the geometry of a transition metal complex, the nature of the ligands, the metal’s oxidation state, and the spectrochemical series influence its optical properties and color?

2 i
15. Which of the following is a co-polymer?
A) Polythene B) Polystyrene
C)PVC ' D)Buna-S-Rubber
16. Which among the following polymers have lowest solubility?
A) epoxy resin B) polyethylene
C) polystyrene D)PET
17. In storage battery plates the matrix commonly used is
A) Aluminium B) Lead
C) Silver D) Copper
18. Which of the following material is commonly used in ceramic matrix composites?
A) Carbon B) Aluminum
C) Silicon carbide D) Rubber
. 19. Kevlar is a
A) Glass B) Thermoplastic
C) Whisker D) Polymer
20. The Bragg’s equation for diffraction of X-rays is.
A)nX = 2d2sin0 B)nk = 2dsin9
QnX = 2dsin20 D)nX=d2sin0
2 1
The potential energy of n-butane is minimum for.
A) Skew conformations B) Staggered conformations
C) Eclipsed conformations . D) Gauche
10. Which of the following stereoisomers will rotate plane-polarized light in opposite directions?
A) Diastereomers B) Enantiomers
C) Geometrical isomers D) Conformational isomers
11. A molecule with a plane of symmetry is.
A) Always optically active ' B) Always a diastereomer
C) Achiral D) Chiral
12. Which of the following is NOTan optically active compound?
A) 1,7- Dicarboxylic Spiro Cycloheptane B) 1,3- Diphenylpropadiene
C) Glyceraldehyde D)meso-tartaric acid

13. The type of linkage present in poly urethane is.
A) Glycosidic linkage B) Phospho diester linkage
C) Ester linkage D) Amide linkage
14. Which of the following is NOT a characteristic of thermosetting polymers?
A) They cannot be remelted after curing B) They are typically brittle and hard
C) They can be reshaped by heating D)They form cross-linked structures during
curing
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'h'
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8 2 4 4
8 2 5 5.
8 .2 5 5
Marks BL CO PO
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27
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22 a. Derive Gibbs-Helmholtz equation of free energy and write any one application.
(OR)
b. Draw a neat sketch of Pourbaix diagram for iron and explain.
23 a. Explain Cahn-Ingold Prelog rules to determine R/S configuration on a chiral center.
(OR)
b. Explain with a neat diagram about the conformational analysis of n-butane.
24 a. Discuss the synthesis, properties and applications of PTFE and Polystyrene.
(OR)
b. Explain n-doping and p-doping in conducting polymers.
25 a. Provide a graphical illustration of the stress-strain relationship in solids and
explain
it in detail.
1 4
(OR)
b. Explain the preparation, properties and uses of Fiber Reinforced Polymer and
Metal Matrix composite.
PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Questions
2 5 5
26 i)Describe with suitable examples, the structural isomerism in coordination
compounds. (8 Marks) .. .

ii) Derive Nemst equation for the redox potential of a reversible reaction and
write its advantages. (7 Marks)
i) Explain the principle and instrumentation of X-ray Photo Electron Spectroscopy.15
(10 Marks)
ii) Discuss in detail about SN1 reaction mechanism. (5 marks)
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